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Study of CeO_2 and Its Native Defects by Density Functional Theory with Repulsive Potential

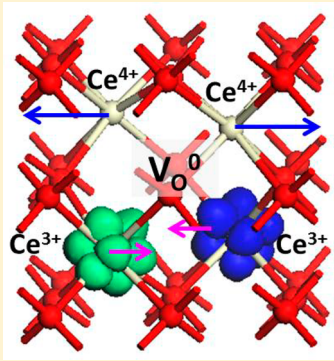
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Supporting Information

ABSTRACT: We investigated the native point defects in CeO_2 by the density functional theory with repulsive potential (DFT+U) method and by use of a nonlinear core-corrected norm-conserving Ce pseudopotential. We find the neutral oxygen vacancy (V_O^0) in CeO_2 to have a very low formation energy of only 0.39 eV in the O-poor limit. It is a deep donor with negative U behavior, stable only in its neutral and doubly positive states. The anion Frenkel defect is found to be the lowest energy disorder defect, with a formation energy of only 2.08 eV/defect site. These low formation energies arise from the improved transferability of our Ce pseudopotential for its +3 and +4 valence states. The negative U behavior of V_O leads to excellent photocatalytic behavior, while the low formation energy of the anion Frenkel defect leads to a superior oxygen storage-and-release capability.



1. INTRODUCTION

CeO_2 is an important lanthanide oxide that is widely used as an oxygen buffer in car exhaust catalysts,[1] as a fast ion conductor in solid state fuel cells,[2] as a catalyst,[3−6] as a high-dielectric constant gate oxide,[7] and in resistance random access memories (ReRAM).[8] Many properties of CeO_2 are determined by its intrinsic defects[9−23] and the unusual behavior of the semicore Ce 4f levels. CeO_2 has mixed-valence behavior accompanied by the localization of electrons on Ce 4f states due to oxygen-related defects.[9−23] The oxygen vacancy is the most common defect of the CeO_2 surface, as noted in both experimental[11] and theoretical studies.[23] It is found that structural distortion plays a key role in stabilizing the vacancy, and this is related to the oxygen storage capacity of CeO_2.[15] On the other hand, the disorder defects are important in oxygen ion transport.[9,10] It is therefore important to understand the properties and energetics of the native defects and in particular the oxygen vacancy and the structural disorder defects.

There have been numerous first-principles studies of the electronic structure of CeO_2.[16−18,21,22,24−29] It is well-known that the conventional density functional theory (DFT) fails to describe well the localized 4f orbitals due to a lack of self-interaction cancellation (SIC) in DFT.[21,22,24−29] There has therefore been considerable effort to correct this error. The simplest method is to use the DFT+U method, which adds an on-site repulsive potential U for the localized d and f orbitals.[30−32] However, this method is somewhat empirical, and further methods such as the GW method[26,27] and the hybrid functionals[23,28,29,33,34] can be used. However, self-consistent GW is computationally expensive. Jiang et al.[26,27] proposed the lower-cost G_0W_0 version, based on an initial electronic structure derived from the local density approximation (LDA) + U method. This gives a substantially improved band structure. Hybrid functionals like the Heyd−Scuseria−Erzenhof (HSE) function or screened exchange (sX) schemes are an efficient way to improve band gaps in electronic structure calculations and to carry out geometry relaxations for defects in these systems.[23,24]

On the other hand, the transferability of pseudopotential is another possible source of error in such mixed-valence systems, and there has been less work on this aspect. The pseudopotential tends to be calculated for a single valence state within the DFT method. It is not necessarily fully transferable. The errors arise because of the spatial overlap of the core and valence charge density. As the exchange−correlation potential is a nonlinear function of the charge density, it is desirable that the core and valence charge densities are spatially separated. Otherwise, the pseudopotential is dependent on the valence configuration and spin polarization. This overlap between valence and core charge density and the nonlinearity causes a systematic error in the total exchange−correlation potential. It is desirable to use a pseudopotential to restrict the basis set for computational speed, especially in defect supercell calculations. However, this source of error can lead to the use of all-electron calculations.

The electronic structure of CeO_2 is still a matter of some debate. It has two gaps between 2p and 4f states and between 2p and 5d states that must be quantified. The experimental data on the band gaps[35−41] mostly have low resolution of 0.3−0.6 eV due to the lifetime broadening. To normalize the experimental values,

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Table 1. Calculated Bulk Properties of CeO_2 Compared with Results of Other Groups and Experiments[a]

	PBE-w/o-NLCC	PBE+U_{pf}+NLCC	ref 21	ref 22	ref 29	exptl data	error vs expt (%)	exptl ref
lattice, a (Å)	5.45	5.46	5.49	5.49	Exp. 5.411	5.411	0.9	54
E_g (2p−4f) (eV)	2.30	3.30	2.51	2.35	4.2	3.33	0.9	40
E_g (2p−5d) (eV)	5.47	5.98	5.30	5.31	6.5	6.0	0.3	39
VB width (eV)	3.90	4.69	4.0	4.0	4.4	4.5−5.0	4.2*	28, 35
formation ΔH_f (eV)	−10.54	−11.50	−10.50	−11.80		−11.28	1.9	56
bulk modulus (GPa)	184	201		181		220	8.6	55

[a]We give the error between our PBE+U_{pf} + NLCC calculated results and experimental values. The asterisk means that we take the value of 4.5 eV as experimental VB width for error comparison.

the most reliable value for the O_{2p}−Ce_{5d} gap is chosen as 6 eV and the O_{2p}−Ce_{4f} gap is taken from the photoluminescence measurements, which show higher accuracy (3.33 and 3.39 eV above the O_{2p} valence band edge).

Here, we carry out DFT+U band calculations using a nonlinear core-corrected norm-conserving pseudopotential, which is a basis for other lanthanide materials. We then carry out calculations on the point defects of CeO_2 as starting point for other lanthanide oxides and the higher actinide oxides with partially filled f orbitals.

2. NONLINEAR CORE CORRECTION FOR 4F LEVELS IN LANTHANIDES

In order to correct the error induced by overlaps of valence-core charge density, we choose the nonlinear core correction (NLCC) method to take the core charge density into account for each atom before computing the LDA or generalized gradient approximation (GGA) potential. For fast Fourier transform, only some of the core charge is included by the partial core correction (PCC). Following Louie et al.,[42] the full core charge density can be replaced with a partial core charge density, defined as below, for nonlinear core correction:

$$\rho^{c}_{\text{partial}}(r) = \begin{cases} A\sin(Br)/r & \text{if } r < r_0 \\ \rho^{c}(r) & \text{if } r > r_0 \end{cases} \quad (1)$$

where r_0 is a radius where the core charge density is 1−2 times larger than the valence charge density. A and B are determined by the value and the gradient of the core charge density at r_0. By this method, the core charge is therefore retained and used to reconstruct the full exchange−correlation potential in the calculations. A similar method has been proposed by Fuchs and Scheffler,[43] but we focus on the theoretical prototype of NLCC used in preliminary calculations.

3. CALCULATION SETUP

Our DFT+U calculation uses the CASTEP[44] code. The norm-conserving pseudopotentials of Ce and O are generated by OPIUM code in the Kleinman-Bylander projector form[45] and already employ nonlinear partial core correction[42] and scalar relativistic averaging scheme[46] for spin−orbital coupling effect. The RRKJ method is chosen as optimization of pseudopotentials.[47] The PBE functional was chosen for PBE+U calculations with a kinetic cutoff energy of 750 eV, which expands the valence electrons states in a plane-wave basis set. The ensemble DFT (EDFT) method of Marzari et al.[48] is used for convergence. For bulk properties of CeO_2, we use a 4 × 4 × 4Monkhost-Pack (MP) k-point mesh, which converges the total energy to under 5.0×10^{-7}eV per atom. The Hellmann−Feynman force on each atom was converged to lower than 0.01 eV/Å.

For defects, we use a 2 × 2 × 2 CeO_2 supercell containing 96 atoms. We select the (1/4, 1/4, 1/4) special k-point[49] in the simple cubic 2 × 2 × 2 supercell. The geometry optimization used the Broyden-Fletcher-Goldfarb-Shannon (BFGS) algorithm through all bulk and defect supercell calculations.

We follow the Anisimov type DFT+U method[30] and the self-consistently determined Hubbard U parameter (U_f = 4 eV) for Ce 4f orbital by ab initio linear response method of Cococcioni et al.[31,32] To stabilize the hole states lying in the O 2p orbitals, we also apply a Hubbard U potential to the O 2p states (U_p = 4 eV) following Lany,[50,51] Morgan et al.,[52] and Keating et al.[21] Accordingly, both the f- and p- orbital electrons of the rare earth and oxygen should be considered when using DFT+U. Note that all of our PBE+U calculations were implemented with NLCC, and U is set to U_f = 4.0 eV for Ce and U_p = 4.0 eV for O, which is U_{pf} for short.

For the calculation of defect formation energy in different charge states, the overall supercell size was kept fixed based on the relaxed neutral bulk unit cell. The defect formation energy (H_q) at the charge state q as a function of the Fermi energy (E_F) and the chemical potential $\Delta\mu$ of element α is given by

$$H_q(E_F, \mu) = [E_q - E_H] + q(E_V + \Delta E_F) + \sum_\alpha n_\alpha(\mu_\alpha^0 + \Delta\mu_\alpha) \quad (2)$$

where E_q and E_H are the total energy of a defect cell and a perfect cell, respectively, calculated of charge q, ΔE_F is the Fermi energy with respect to the valence band maximum, n_α is the number of atoms of element α, and μ_α^0 is reference chemical potential, following Lany and Zunger.[53]

4. RESULTS AND DISCUSSION

4.1. Bulk. Bulk CeO_2 has the cubic fluorite structure, with lattice constant 5.411 Å.[54] Table 1 shows our Perdew−Burke−Ernzerhof (PBE) result from geometry optimization is 5.450 Å, with an error of 0.7%. The PBE+U calculation similarly gives 5.458 Å with around 0.9% error, representing comparable reliability when the U parameter is introduced. The PBE+U results compare with the 5.494 Å value found by Keating et al.[21] and Zacherle et al.[22] within the GGA+U scheme. The bulk modulus of CeO_2 is calculated by PBE and PBE+U and the results obtained are B_0 = 184 GPa for PBE and B_0 = 201 GPa for PBE+U. Though these values are smaller than the experimental value of 220 GPa,[55] they are improved over the values [B_0 = 172 GPa (GGA) and 181 GPa (GGA+U)] found by Zacherle et al.[22] The calculated electronic density of states (DOS) with PBE+U (U_f = 4, U_p = 4) under NLCC has good agreement with X-ray photoelectron spectroscopic (XPS) experimental data; see Figure 1.

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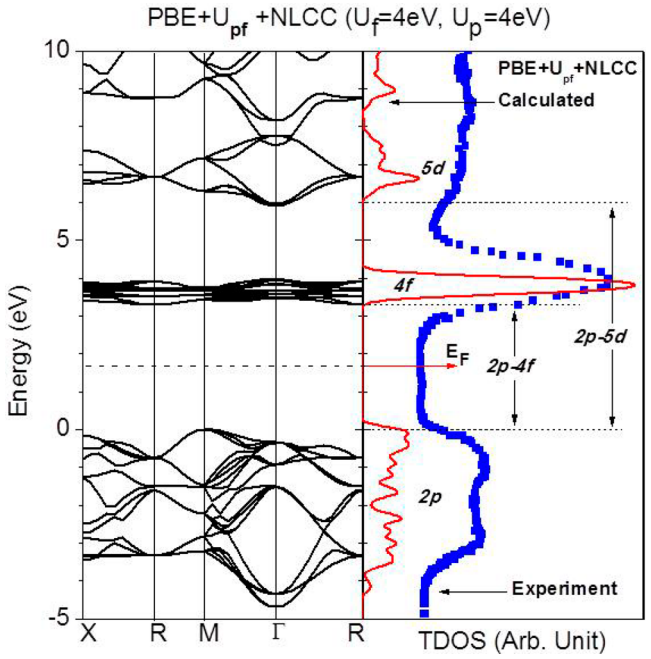


Figure 1. (Left) Calculated band structures of CeO_2 by PBE+U with U_f = 4 eV and U_p = 4 eV based on NLCC. The black dashed line is the Fermi level (E_F). (Right) Calculated total density of states of CeO_2 by nonlinear core corrected PBE+U with U_f = 4 eV and U_p = 4 eV (shown as PBE+U_{pf} + NLCC; red solid line), where the 2p−4f gap is 3.30 eV and the 2p−5d gap is 5.98 eV. The zero energy is the highest occupied level. The blue dotted line denotes the experimental results reprinted and replotted from literature results.[35] The red arrow shows the Fermi level (E_F).

The band structures of CeO_2 calculated by both GGA and GGA+U methods (see Figure 1) are compared. This work found that the band gap of O_{2p}−Ce_{4f} orbitals increases from 2.30 eV in GGA to 3.30 eV in GGA+U, which is very consistent with the experimental value of 3.33 eV from photoluminescence.[40] The secondary band gap of O_{2p}−Ce_{5d} orbitals also increases from 5.47 eV in GGA to 5.98 eV in GGA+U, which is close to the optical reflectance value of 6.0 eV.[39] The upper valence band (mainly 2p orbitals of oxygen atoms) width increases from 3.90 eV in PBE to 4.69 eV in GGA+U, which matches the range of 4.5−5.0 eV reported by Wuilloud et al.[35] and Mullins et al.[37]

For the bulk formation enthalpy, $\mu_{Ce} + 2\mu_O = \Delta H_f(CeO_2)$, the experimental formation enthalpy of CeO_2 is −11.28 eV at T = 298 K.[56] Our calculations give −10.54 eV by PBE and −11.50 eV by PBE+U (U_f = 4, U_p = 4) at T = 0 K (ground state), which agrees well with the experimental value at T = 298 K.[46] Before we discuss the defect formation energy, the value of the chemical potential of oxygen and cerium in CeO_2 is determined by the limitation of the phase boundaries between CeO_2 and Ce_2O_3. The upper bound of μ_O, which is the Ce-poor/O-rich environment, is limited by the formation of O_2 molecule (gas), which means μ_O = 0 eV and μ_{Ce} = −11.50 eV. The lower bound of μ_O, the Ce-rich/O-poor limit, is given by the formation of Ce_2O_3, or $2\mu_{Ce} + 3\mu_O \le \Delta H_f(Ce_2O_3)$ = −20.23 eV, giving μ_O = −2.77 eVand μ_{Ce} = −5.95 eV.

4.2. Oxygen Vacancy. The neutral oxygen vacancy leaves two excess electrons, which localize onto two of four adjacent Ce sites, changing their valence from Ce^{4+} to Ce^{3+}. For V_O^0, after geometry relaxation, eight neighboring oxygen atoms move off their ideal lattice sites toward the vacancy site since a vacancy site is an effective positive charge center, while neighboring Ce atoms move away from the vacancy with the two Ce^{4+} moving further.

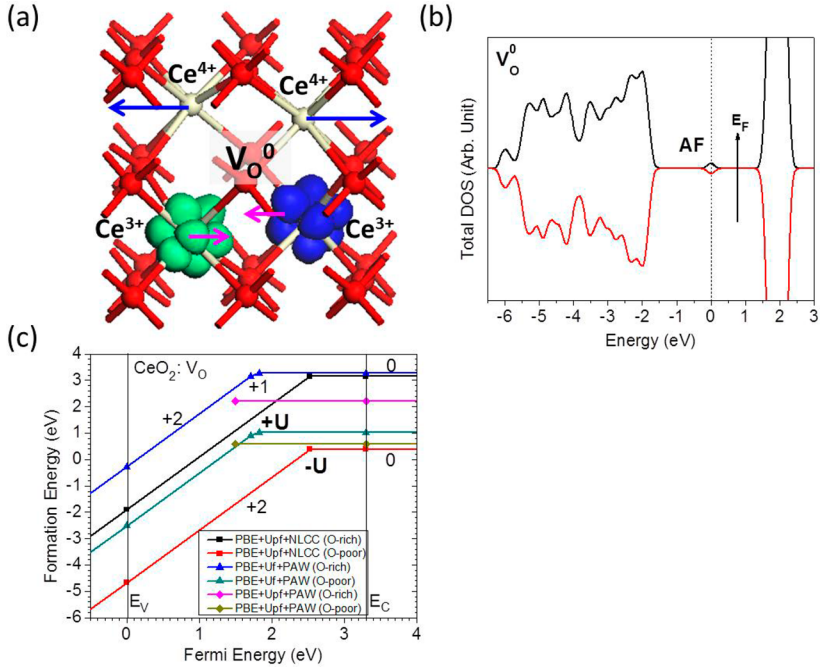


Figure 2. (a) Local view of relaxed structure of CeO_2 with V_O by PBE+U. Blue and green isosurfaces denotes the two electrons localized on the f orbitals of two Ce atoms and therefore producing gap states with spin-up and spin-down states, respectively. Ce^{3+} and Ce^{4+} are shown. The length of the arrows indicates the distance of the different displacements of Ce ions. (b) Total DOS of CeO_2 with V_O in neutral state. The Fermi level is shown by the arrow. AF means antiferromagnetic configurations of the defect states. (c) Summary of calculations on formation energies of V_O, calculated by PBE+U_{pf} with NLCC (this work), PBE+U_{pf} with projector augmented wave (PAW),[21] and PBE+U_f with PAW[22] under both O-poor and O-rich chemical potential limits.

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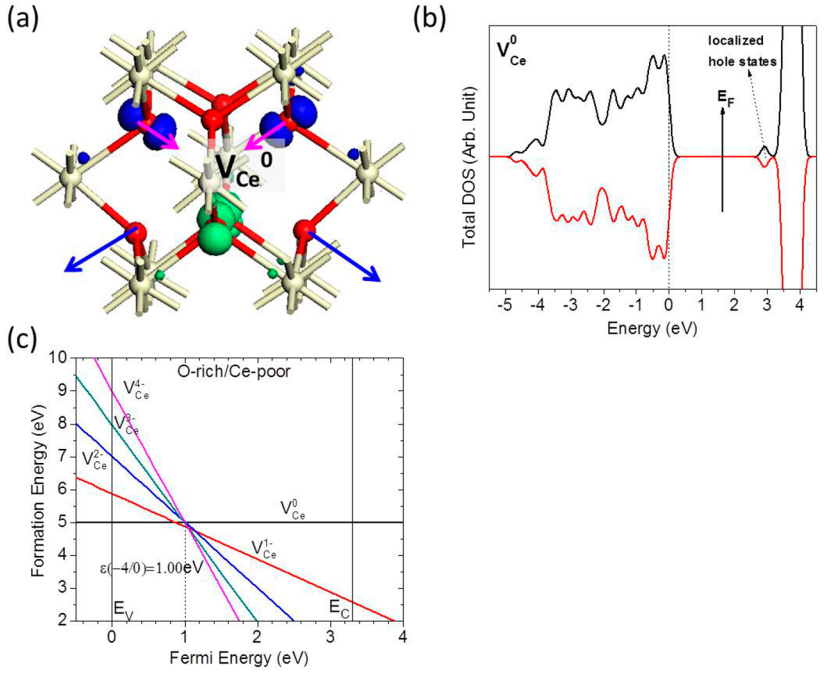


Figure 3. (a) Local side view of relaxed structure of CeO_2 with V_{Ce} (located in the center). Blue and green isosurfaces denotes the four holes localized on the nearest O sites: two of them are spin-up and the other two are spin-down states, respectively. The length of the arrows indicates the distance of the different displacements of O ions. (b) Total DOS of CeO_2 with V_{Ce} in neutral state. The vertical dashed line at zero energy denotes the highest occupied level. The Fermi level is shown by the arrow. (c) Formation energy of V_{Ce} in charge states 0, −1, −2, −3, and −4 under the O-rich (or Ce-poor) chemical potential limit.

Moreover, the neighboring O atom bridging two Ce^{4+} moves by 0.25 Å, while the O atom bridging two Ce^{3+} moves only 0.01 Å from our PBE+U calculations. The four O atoms connected to one Ce^{3+} move toward V_O^0 by 0.15 Å. The difference of O atoms is due to Ce^{3+} ions having less positive charge than Ce^{4+}. The two Ce^{3+} ions move a distance of 0.13 Å toward each other due to weak coupling of two unpaired electrons in 4f states, while the two Ce^{4+} ions move a distance of 0.18 Å (Figure 2a). This trend is absent in simple PBE calculations.

Figure 2b shows the electronic density of states (DOS) of CeO_2 with V_O^0 calculated by PBE+U (U_f = 4 eV, U_p = 4 eV). It can be seen that O_{2p} dominates the valence band and Ce_{4f} dominates the conduction band as the 4f band is empty. Between O_{2p} and Ce_{4f} bands, there are two gap states, spin-up and spin-down, lying well inside the band gap (near the midgap) with antiferromagnetic arrangement. Moreover, these defect states denote the occupied Ce_{4f} states, indicating the occurrence of Ce^{3+} ions near the V_O site.

For the DOS of V_O^+, there is only a gap state with a single peak near the midgap. The gap states of V_O^0 are 1.55 eV below the conduction band edge. The gap state of V_O^+ also lies near midgap, at about 1.05 eV below the conduction band edge.

Figure 2c shows the defect formation energies of oxygen vacancy for both O-poor and O-rich conditions within different charge states by PBE+U (U_f = 4 eV, U_p = 4 eV). For V_O^0 under O-rich conditions, the calculated formation energy is 3.16 eV, corresponding to 0.39 eV for the O-poor limit. The formation of V_O^0 under O-rich conditions is higher than the value of 2.22 eV by Keating et al.[21] but closer to 3.27 eV by Zacherle et al.,[22] while our V_O^0 formation energy of 0.39 eV in the O-poor limit is lower than both 1.03 eV of Zacherle et al.[22] and 0.58 eV of Keating et al.[21] The transition energy levels correspond to Fermi energies where two charge states, q and q', have the same formation energies according to eq 2. The calculated transition energy level of V_O^0/V_O^{2+} by PBE+U (U_f = 4 eV, U_p = 4 eV) is 2.53 eV, which is 0.77 eV below the conduction band minimum E_C. This compares to ∼1.7 eV by Zacherle et al.,[22] 2.0 eV by Jiang et al.,[19] and 1.5 eV by Hellman et al.[20] The 0/+ transition energy level of V_O is 0.97 eV below E_C. However, V_O^+ is never the most stable charge state. The defect tends to disproportionate according to the reaction $2V_O^+ \rightarrow V_O^0 + V_O^{2+}$, leaving a neutral ($V_O^0$) and a +2 charge state (V_O^{2+}). This corresponds to a so-called negative U defect, with U = −0.39 eV. The negative-U behavior of V_O in CeO_2 is in accordance with the experimental phenomenon that catalytic activity of CeO_2 with V_O is higher than the perfect bulk CeO_2. Thus, the oxygen vacancy is a near-deep donor defect, of which V_O^0 and V_O^{2+} are the stable states.

There has been a long debate on the ferromagnetic (FM) behavior that was observed in nonmagnetic oxides, when synthesized as thin film or as nanoparticles. However, CeO_2 is unusual. Castleton et al.[41] and Andersson et al.[17] both found that the antiferromagnetic (AFM) and FM configurations are degenerate with ΔE < 1 meV in the bulk CeO_2 with V_O. The reduced CeO_2 surface has similar effects. Fabris et al.[57] found that V_O causes CeO_2 surface to have AFM behavior independent of the V_O concentration, while Nolan et al.[58] and Han et al.[59] noted that oxygen vacancies will induce FM in CeO_2.

Here, we find AFM behavior of the neutral state of V_O in CeO_2 and near-degeneracy of the FM state. This is similar to the results obtained by Keating et al.,[21] Andersson et al.,[17] and Castleton et al.[41] Zhang et al.[60] also find that AFM is slightly more stable than FM by 0.02 eV.

4.3. Cerium Vacancy. The cerium vacancy (V_{Ce}) gives four hole states on its adjacent O sites. Figure 3a shows the relaxed structure of V_{Ce} with four localized hole states. The defect has a quintet spin state that is slightly lower in energy than the triplet or singlet states and also selectively localized on specific O atoms with T_d symmetry. For the local structure relaxed by PBE+U (U_f

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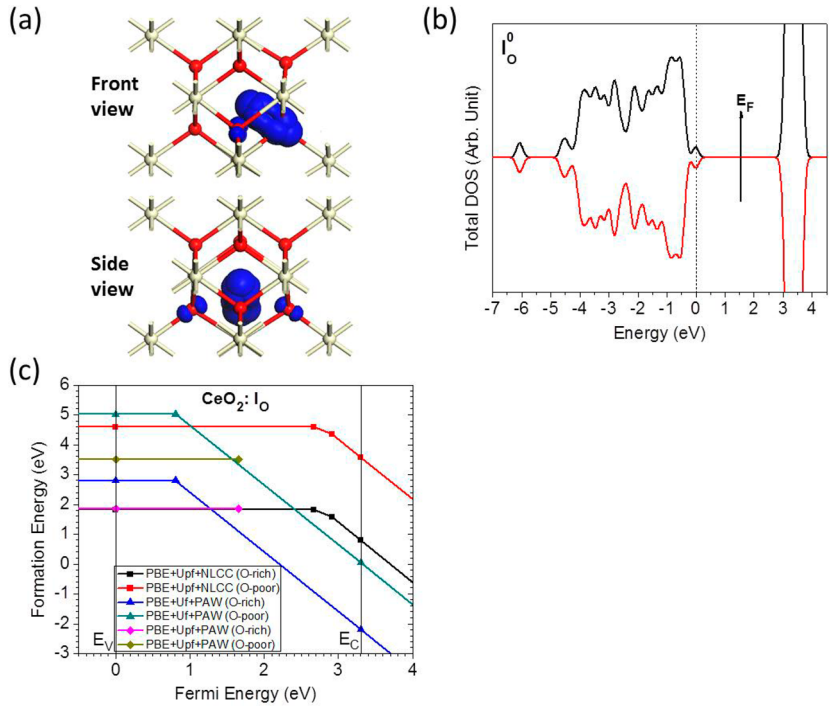


Figure 4. (a) Local front and side views of relaxed structure of CeO_2 with I_O. The localized orbitals are sitting along I_O and host O ions. Small π-orbitals locate the O ions next to I_O. (b) Total DOS of CeO_2 with I_O. The vertical dashed line at zero energy denotes the highest occupied level. The Fermi level is shown by the arrow. (c) Summary of calculations on formation energies of I_O in CeO_2 calculated by PBE+U_{pf} with NLCC (this work), PBE+U_{pf} with PAW,[21] and PBE+U_f with PAW[22] under both O-poor and O-rich chemical potential limits.

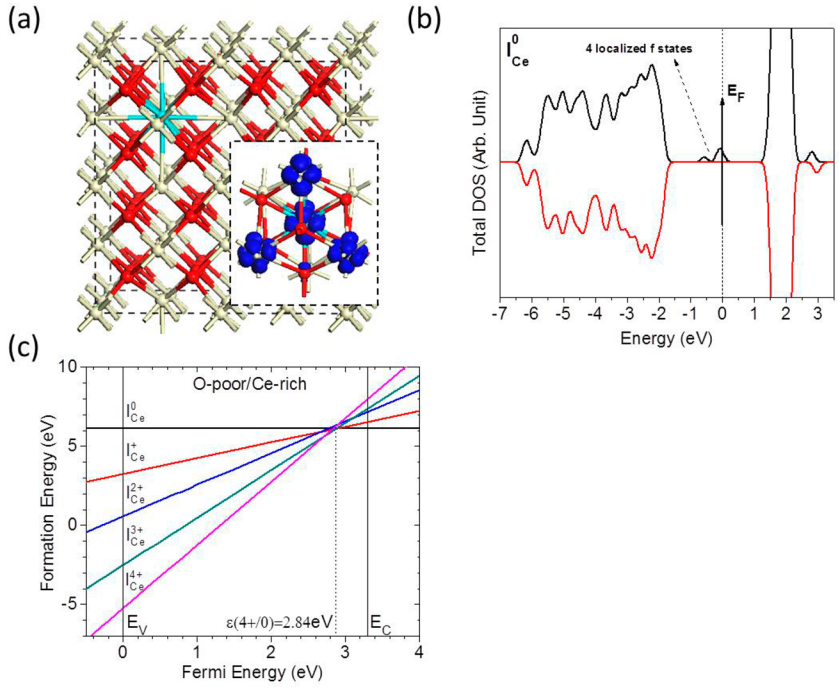


Figure 5. (a) Relaxed structure of I_{Ce} in CeO_2. The light blue sphere is the I_{Ce} atom. The right bottom figure is the local view of localized gap states orbitals induced by I_{Ce} in CeO_2. (b) Total DOS of CeO_2 with I_{Ce}. The vertical dashed line at zero energy denotes the highest occupied level. The Fermi level is shown by the arrow. (c) Formation energy of I_{Ce} in CeO_2 under O-poor (or Ce-rich) chemical potential limit.

= 4 eV, U_p = 4 eV), the O atoms without four localized hole states uniformly move away from the V_{Ce} site by 0.35 Å, whereas the O atoms with localized hole states all move toward the V_{Ce} site by 0.10 Å. This phenomenon cannot be found by simple PBE+U without U parameters on O_{2p}. Our results on local O atoms movement are all larger and more obvious than the results of Keating et al.[21]

From the electronic DOS of CeO_2 with V_{Ce}, shown in Figure 3b, the system shows a defect state with 0.55 eV below the conduction band minimum (CBM), which shows that the

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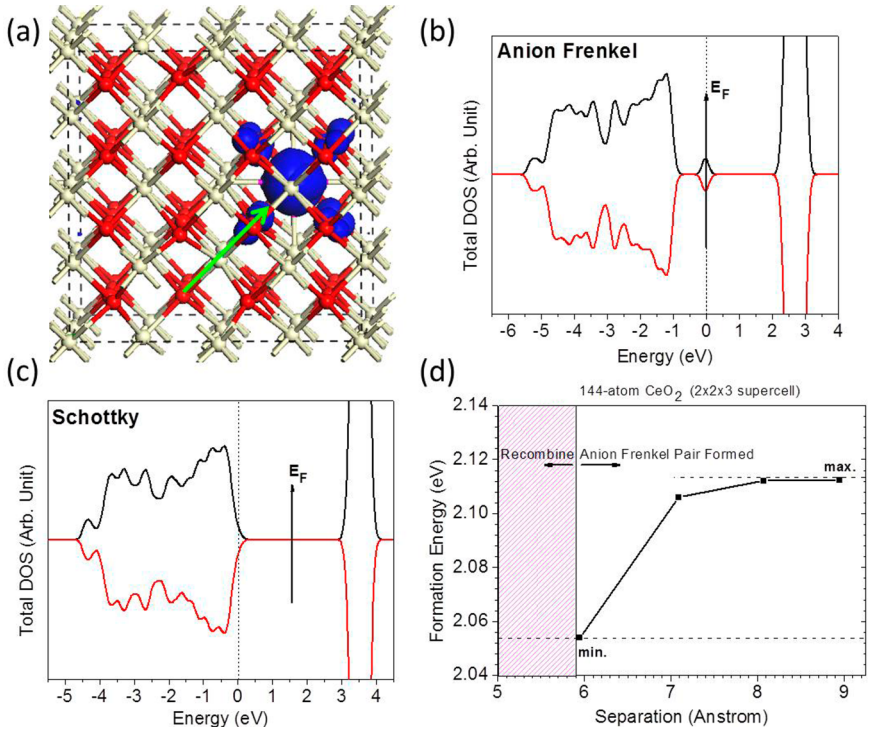


Figure 6. (a) Relaxed structure of anion Frenkel defect ($V_O + I_O$) in CeO_2. The green arrow shows the formation path of the anion Frenkel defect in CeO_2. The starting point of the arrow is the oxygen vacancy (VO) site. The arrow points to the oxygen interstitial (I_O) site that located at the centre of the localized orbitals (blue isosurface area). Localized orbitals of the gap states induced by the anion Frenkel defect ($V_O + I_O$) in CeO_2. (b) Total DOS of the anion Frenkel defect ($V_O + I_O$) in CeO_2. The vertical dashed line at zero energy denotes the highest occupied level. The Fermi level is shown by the arrow. (c) Total DOS of the Schottky defect ($V_{Ce} + 2V_O$) in CeO_2. The vertical dashed line at zero energy denotes the highest occupied level. The Fermi-level is shown by the arrow. (d) Formation energies (per defect) of anion Frenkel (a-Fr) pair defect with reference to separation (in angstroms) of the a-Fr pair.

localized hole states are dominated by the empty O_{2p} orbitals. The calculation by PBE+U (U_f = 4.0 eV, U_p = 4.0 eV) shows the importance of including a U interaction of the O 2p states, otherwise V_{Ce} in CeO_2 would have a metallic DOS, due to the incorrect description of the hole states.

Calculated formation energies of cerium vacancy are shown in Figure 3c with different charge states of V_{Ce}^{0}, V_{Ce}^{-1}, V_{Ce}^{-2}, V_{Ce}^{-3}, and V_{Ce}^{-4}. The transition energy level of 0/−4 is 1.00 eV above the valence band maximum (VBM). The formation energy of V_{Ce}^{0} is 5.00 eV for the O-rich limit and 10.55 eV for Ce-rich. Compared to Zacherle et al.,[22] our results gave values 1.22 and 0.15 eV lower under O-rich and O-poor limits, respectively. On the other hand, Keating et al.[21] show very similar results under the O-rich limit, which is 4.91 eV with only 0.09 eV difference compared with ours. However, their formation energy of V_{Ce}^{0} in the O-poor limit is remarkably low (8.19 eV), about 2.5 eV lower than the results of Zacherle et al.[22] and ours. This may due to overestimation of μ_{Ce} with larger U_f and U_p values.

4.4. Oxygen Interstitial. The local structural relaxation between oxygen interstitial (I_O) and lattice oxygen sites leads to absence of hole states. The relaxed structure of the oxygen interstitial (I_O) is obtained by PBE+U with U_f = 4.0 eV and U_p = 4.0 eV. I_O is found to re-bond to an adjacent lattice O site with a bond length of 1.391 Å. This is larger than the O−O bond length (1.217 Å) of O_2 and shows characteristics of a peroxide species. The lattice O site moves 0.071 Å toward the I_O, and the I_O moves 0.88 Å toward the neighboring lattice O site. Figure 4a shows the localized orbitals around the oxygen interstitial site. The oxygen interstitial forming peroxide ion in closed packed metal oxides is also seen in Al_2O_3, TiO_2, and ZnO.

In the electronic DOS of I_O (Figure 4b), there are three peaks with both spin-up and spin-down, which lie in the range from −6.5 to 0 eV. The σ state of the peroxide is at −4.4 eV, slightly below the lower limit of the valence band. The π and π^* states are found at −6.05 and 0 eV, respectively.

The formation energy of I_O in charge states of 0, −1, and −2 under O-rich chemical potential limit are 1.82, 4.49, and 7.40 eV, respectively. The formation energy of I_O^{0} is about 1 eV lower than the value obtained by Zacherle et al.[22] As seen from Figure 4c, I_O gives almost the same formation energy under O-rich conditions as found by Keating et al.,[21] which supports the reliability of introducing U on O_{2p}. We also found the positive-U effect between 0 and −2 states compared to −1 state, showing that the energy cost of merely introducing O cannot be fully compensated by disordering the lattice.

4.5. Cerium Interstitial. Interstitial Ce atom provide four excess electrons (Figure 5a) with one electron remains on the interstitial Ce (I_{Ce}) while the three others move to the neighboring cerium atoms (see right bottom of Figure 5a). The local structure relaxed by the PBE+U method with U_f = 4.0 eV and U_p = 4.0 eV shows that three of the nearest-neighboring Ce atoms move 0.244 Å away from their original Ce lattice positions, while the remaining three nearest-neighboring Ce atoms move 0.324 Å away from their original positions. Analyzing the orbital composition, we find that the three Ce atoms with 0.244 Å movement are the Ce^{3+} ions and the other three Ce atoms with 0.324 Å movement are the Ce^{4+} ions.

From the electronic DOS results shown in Figure 5b, the ground-state electronic configuration of the CeO_2 system with I_{Ce} presents a ferromagnetic feature. There are in fact four states

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deep in the gap, and two states have nearly the same energy pinning of the charge neutral level (CNL) 1.45 eV below the CBM. These deep levels stay 1.45, 1.53, 1.57, and 2.03 eV below the CBM with the same spin-up configurations and fully occupied states, which is different from the results of Keating et al.,[21] who reported the system as being antiferromagnetic and all defect states as being partially occupied.

The energy cost of accepting the additional Ce in CeO_2 as I_{Ce} is very high in the neutral state: 11.69 eV for I_{Ce}^{0} at the Ce-poor (or O-rich) limit and 6.14 eV at the Ce-rich (or O-poor) limit, shown in Figure 5c. The formation energy of our I_{Ce}^{0} is around 1 eV higher than those reported by Zacherle et al.[22] due to the difference of processing the chemical potential with different pseudopotentials and Hubbard U parameters. However, a large difference is presented for I_{Ce}^{4+} at the Ce-poor limit since the formation energy of I_{Ce}^{4+} of 5.60 eV by Zacherle et al.[22] is about 5 eV higher than ours. This may be due to errors when calculating the ionization energies with different pseudopotentials and U parameters.

4.6. Frenkel and Schottky Defects. We have also studied the anion Frenkel and Schottky defects, respectively. These defects are important, as they determine how CeO_2 responds to disorder. The anion Frenkel defect corresponds to moving an oxygen atom from its original lattice site to leave an oxygen vacancy (V_O) and create an interstitial (I_O) (see Figure 6a). From the electronic DOS calculation, it is found that there is an acceptor-like state appearing at 1.02 eV above VBM (see Figure 6b). The orbitals of the defect state are mainly dominated by π orbitals of I_O while there are no states from V_O. From the relaxed structure, the lattice around V_O is more obviously distorted than the structure near I_O, since the V_O localizes two electrons in the f orbitals of adjacent Ce atoms, which cause large Coulomb repulsive interactions, while the π-orbitals around I_O from the oxygen atoms will participate in new coupling interactions and therefore induce less lattice distortion. Meanwhile the excess O_{2p} level of I_O will shift to a higher energy due to the Coulomb repulsive interaction.

In the case of the Schottky defect, the structure has not been affected by the presence of the defect. Also, no gap states have been found in the electronic DOS (Figure 6c), since four holes generated by V_{Ce} passivate the four electrons left by two V_O.

We found that the anion Frenkel defect can easily form with a relatively low formation energy of 2.08 eV, while the Schottky defect cost more energy (3.86 eV). Note that the anion Frenkel defect is generated through formation of V_O and I_O within diffusion distance. We recall that V_O has a low formation energy of 0.39 eV in the O-poor limit. Therefore, we believe that the V_O + I_O complex will have a similar formation energy to the anion Frenkel defect under specific charge state and chemical potential environment. This implies the possible dynamics of oxygen-assisted defects (vacancy formation, migration, etc.) in the CeO_2 system, which would reveal the physical nature of oxygen storage and reversible catalytic behaviors.

The anion Frenkel (a-Fr) defect is the dominant defect contributing to the ionic conductivity and diffusion of CeO_2. We find it to be the lowest-cost disorder defect (2.08 eV in 96-atom supercell), which is consistent with CaF_2 or other materials in fluorite structure. Table 2 shows that our value of 2.08 eV is slightly higher than that found by Keating et al.[21] (1.98 eV) and similar to that of Zacherle et al.[22] (2.07 eV). Thus, the anion Frenkel defect will account for the high ionic conductivity of CeO_2, consistent with the experimental data of Stratton and Tuller.[10]

Table 2. Formation Energy Comparison for Neutral Oxygen Vacancy (V_O) Anion Frenkel, Frenkel, and Schottky Defects[a]

defect	ref 22	ref 21	this work
neutral V_O (O-poor), eV	1.03	0.58	0.39
anion Frenkel, eV	2.07	1.98	2.08
Frenkel, eV	6.23		5.80
Schottky, eV	2.29	3.66	3.86

[a]As denoted from the table, this work shows the improvement on the defect formation energy calculation by nonlinear core correction on Ce pseudopotentials.

We have calculated the effect of defect pair separation on the formation energy of a-Fr pair. The formation energy converges to 2.03 eV for large supercell sizes (Figure S1 in Supporting Information). Thus, the 2 × 2 × 3 supercell of 144 atoms is a suitable size to model the a-Fr pair diffusions. Figure 6d also shows that 5.95 Å represents a crossover separation; below this the a-Fr pair recombines, whereas above this the pair can remain separate, due to the stabilization of separate defects by lattice distortion.

With considerable effort given to verifying the results, our value for its formation energy indeed shows less error than that given by Keating et al.[21] because we included both the U_p term on the O 2p orbitals and a more transferable Ce pseudopotential to represent the energy difference between Ce +3 and +4 states.

5. CONCLUSION

This work has used a corrected DFT+U method combined with partial core correction (under nonlinear core correction scheme) in the pseudopotentials, which can ameliorate the conventional DFT+U deficit and is remarkably close to the reported experimental results. We have used this for a systematic study of the native point defects in the CeO_2 system. We find that the V_O has a low formation energy of 0.39 eV in the O-poor limit, accounting for the ease of reduction, and that the anion Frenkel defect is the lowest-energy disorder defect, consistent with experiment (ref 37 of Zacherle et al.[22]). This shows the importance both of including U correction on Ce 4f and O 2p states and of having a fully transferable pseudopotential.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

One figure showing minimum formation energies (per defect) of a-Fr pair defect with respect to supercell size of the a-Fr pair formed. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

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